Group 22

CSE 515: Advanced Communication Networks and System

A

Project Report

on

INVESTIGATION OF VARIANTS OF LEACH PROTOCOL IN WIRELESS SENSOR *NETWORKS*

Dhiraj Pandey – 24091005(ECE)

INTRODUCTION

The wireless sensor network is combination of small devices called sensor nodes. These small devices are deployed in field are used to monitor physical phenomena such as temperature, humidity and vibration [1]. The sensors are divided into three parts: a sensing unit for a computing unit for data processing and storage, and a wireless communication unit for data transmission. Sensor nodes also need power to complete their tasks. The energy used by the sensor node is a low-energy battery because the node can be deployed anywhere and in environments where it is difficult to replace the battery, energy has become the focus of our network [2].

Low Energy Adaptive Cluster Hierarchy (LEACH) is one of the earliest WSN cluster-based protocols. If the LEACH comes from the cluster head used to communicate with the receiver node or base station, direct communication with the receiver node can be avoided, thereby reducing power consumption in the LEACH protocol [3].

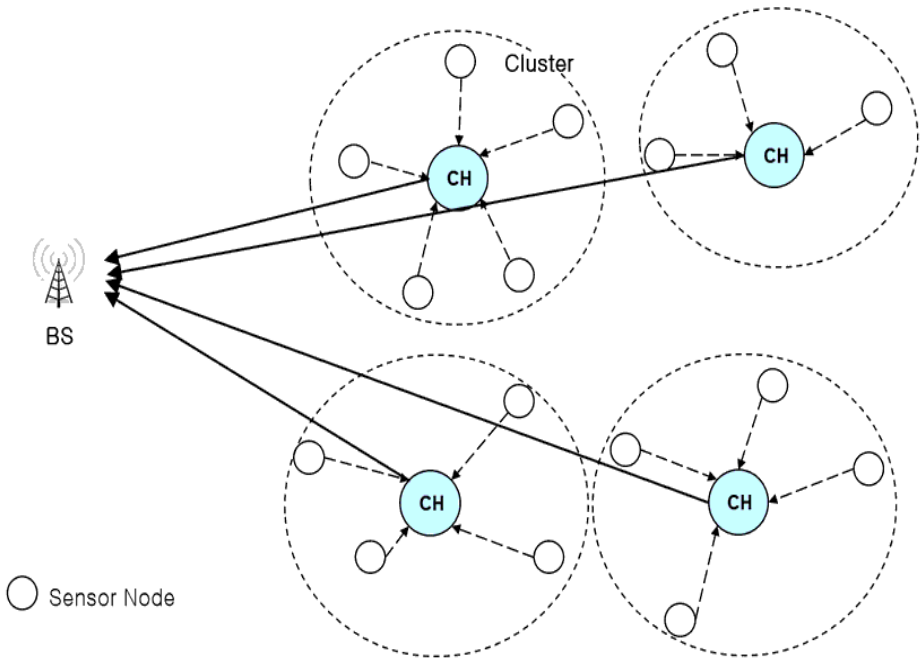


Figure 1.1: LEACH Protocol

LEACH operation is divided into several rounds, similar to the beacon interval concept in the MAC layer, each round has two stages:

The configuration phase is used to form a group, and during the steady-state phase of data aggregation and data transmission to the sink node. In the configuration phase, the cluster head will be selected before the clusters formed, and the cluster will be selected according to a threshold between 0 and 1. if this random number is less than this value, the node will become the cluster head of the node This round [4].

The threshold is defined as:

$$T(n) = \begin{cases} \dfrac{P}{1 - P * (r \bmod \frac{1}{P})} & if\ n \in G \\ 0 & otherwise \end{cases}$$

In the above equation, P is the percentage of the required group head, r is the count of the current round, and G is the set of sensor nodes that were not selected as the group head in the most recent round /P. Define the cluster head, and send the advertisement message to all the remaining sensor nodes with the same transmission power. The remaining sensor nodes decide to join the cluster head according to the maximum signal strength received from this round of announcements [4].

After completing the setup phase, the stabilization phase will last longer than the setup phase to avoid unnecessary burden on group formation. In the stabilization phase, the sensor node will perform the detection and send the data to the cluster head, which will add the message and send it to the base station. The sensor node radio can be turned off to save power until a predetermined transmission time is reached, instead, the cluster head must always keep the radio on receiver data. Before sending data to the coordinator, the cluster manager performs the necessary data processing, namely aggregation and compression. Once this phase is completed, WSN will enter the next round of LEACH protocol and repeat the same configuration and steady-state phase described above [5].

1.1 Sensor Node Structure

A sensor node typically composed of four main components as shown in figure 1.2

1-**Sensig Unit**: The sensing unit usually consists of one or more sensors and analog to digital converters (ADCs). The sensors observe the physical phenomenon and generate analog signals based on the observed phenomenon. The ADCs convert the analog signals into digital signals, which are then fed to the processing unit [2].

2-Processing Unit: The processing unit usually consists of a microcontroller or microprocessor with memory, which provides intelligent control to the sensor node [2].

3-Communication Unit: The communication unit consists of a short-range radio for performing data transmission and reception over a radio channel [2].

4-Power Unit: The power unit consists of a battery for supplying power to drive all other components in the system. In addition, a sensor node can also be equipped with some other units, depending on specific applications [2].

A. Advantage of LEACH

1. Increase the network lifetime.
2. Energy consumption.
3. Allocation of TDMA schedule.
4. Randomized rotation of CH enhances the network lifetime [6] [7].

B. Disadvantage of LEACH

1. In each round the CH is chosen randomly and the probability of becoming the CH is the same for each sensor node. Therefore, robustness of the network is affected and lifetime of the network degrades.
2. LEACH does not guarantee the position and number of CHs in each round. Formation of clusters in basic LEACH is random and leads to unequal distribution of clusters in the network.
3. LEACH follows single hop communication between the CH and the BS [7].

1.2 Objectives

To implement and simulate a wireless sensor networks using MATLAB simulator.

Develop a new technique to increase the lifetime for wireless sensor networks.

To increase the lifetime of wireless sensor networks.

1.3 Motivation

Save the energy of cluster head to increase the lifetime of the WSN. In a LEACH protocol, cluster head sends query to all nodes in its cluster, aggregates data from all the nodes and report it to the base station. The cluster head spends more energy than other nodes in the network, so its energy decreases sharply. The node that has less energy cannot become a cluster head. The goal is how to save energy a cluster head so that the network lifetime increases.

1.4 Tools

Simulation environment is considered for the implementation of WSN as part of this work. MATLAB selected due to its ability to study the effect of different parameters on the system behaviour. MATLAB /SIMULINK used as the tool to build the simulation environment, Advantage and the appropriate of using environment to conduct the simulation are fast and easy prototyping in MATLAB, testing in dynamically changing environment, easy debugging, provide visualization [8]. The other advantage of this method is its flexible in building the end nodes and sensors. This simulation methodology used to build different WSN types and opens the doors to use the MATLAB in new field. MATLAB provides an easy way of application prototyping with visualization capabilities [8].

LITRATURE REVIEW

This part investigates of exiting study which is relevant to our theme and present some background reading required Sharma, M., & Shaw, A. K, (2012). [9] The scope of this paper is the protocol assisted LEACH which achieves lessened and uniform distribution of dissipated energy by separating the tasks of routing and data aggregation. It introduces the concept of helper node which assists cluster heads for multi-hop routing. A new algorithm has been formulated to facilitate energy efficient multi-hop route for helper nodes to reach base station. The proposed protocol extends the lifetime of the networks, minimizes overall energy dissipation in the network and distributes dissipation among cluster heads, sensor nodes and helper nodes on LEACH [Aslam, M., et al. 2012][3] The main focus of his study is how to extend routing protocol work in order to increase the lifetime, and how quality routing protocol is improved for the wireless sensor network. In hierarchical routing protocols whole network is divided into multiple clusters. One node in each cluster play leading rule. Cluster head is the only node that can communicate to base station in clustering routing protocols. This significantly reduces the routing overhead of normal nodes because normal nodes have to transmit to cluster-head only.

[Vashisht & Khurana.2013] The scope of this paper includes both the first and the second levels. The global problem of maximizing network lifetime under the joint clustering, routing, and coverage constraint. Many others address integrated problem of area coverage and network connectivity but they do it in flat networks and don't reap the benefit the energy saving and ease of manageability of cluster-based networks.

Various methods for minimizing energy consumption in wireless sensor network have been proposed such as by Heintzelman et al. [8] who described the LEACH protocol as a hierarchical self-organized cluster-based approach for monitoring application. The data collection area of the data is randomly divided into clusters. Leach uses time division multiple access (TDMA), To transmit data from the sensor nodes to the cluster head. Then CH aggregates the data and transmits it to the base station for processing. One of the features of LEACH is localized coordination and control for the formation and operation of clusters. The cluster head rotate randomly.

In [1] Lindsey et al. came about the proposition of PEGASIS which is an extension of LEACH. It eliminates the overhead of dynamic cluster formation created by LEACH. In this

protocol, the nodes transmit to the CH and transmission of data is done by the cluster head, which is selected in a rotational manner, to the BS, PEGASIS protocol is found to save more energy and is more robust in node failure when compared to LEACH.

[5] Behrouz, R. (2002) developed a protocol that creates clusters of the similar size and uses multi-hop routing between CH and BS. The cluster head which forward the last hop is selected randomly from the sets of cluster heads to minimize the load of cluster head which are located nearest to the base station.

In [8], Wei Li proposed a geometric programming model to extend the network lifetime of the sensor network by clustering sensor nodes into groups. He developed an iterative method for solving the geometric programming by choosing the optimal location of cluster heads. The optimum mentioned in his proposition refers to minimizing energy consumption based on to inter-sensor network under specific constrained.

Clustering of approaches is useful in the monitoring of habitat and environs. This however, necessitates the use of continuous stream of sensor data. Xinhua Liu et al. [6] propose DDBC (Directed Diffusion Based on Clustering). DDBC is an energy-efficient directed diffusion routing protocol which is based on the reduction of the network topology and gives suppression to the redundancy message in plain flooding in order to minimize energy consumption in wireless sensor network.

Ye, Heidemann and Estrin [7] gave a description of a contention based medium access protocol, S-MAC, which minimizes energy consumption in wireless sensor network by using virtual clusters. They developed the common sleep schedule for the clusters and overhearing is avoided by the use of in-channel signaling.

Wei Cheng et al. [18] proposed a novel adaptive, distributed, energy efficient clustering algorithm, AEEC for wireless sensor network. Their approach selects cluster heads based on the node energy related to that of the whole network which can bring about efficiency in heterogonous networks.

METHODOLOGY USED

MATLAB is a powerful language for technical computing, the name MATLAB stands for Matrix Laboratory, because its basic data elements are a matrix (Array), MATLAB can be used for math computations, modelling and simulation, data analysis and processing, visualization and graphic [10].

3.1 Energy model of Sensor Nodes (First Order Radio Model)

The radio characteristics, including energy dissipation in the transmit and receive modes. Simple model where the radio dissipates Eelec = 50nJ/bit to run the transmitter or receiver circuitry and Eamp = 100 pJ/bit/m^2for transmit amplifier. Transmit a k-bit message a distance d, assume k = 4000 data/bit. The radio expends [1];

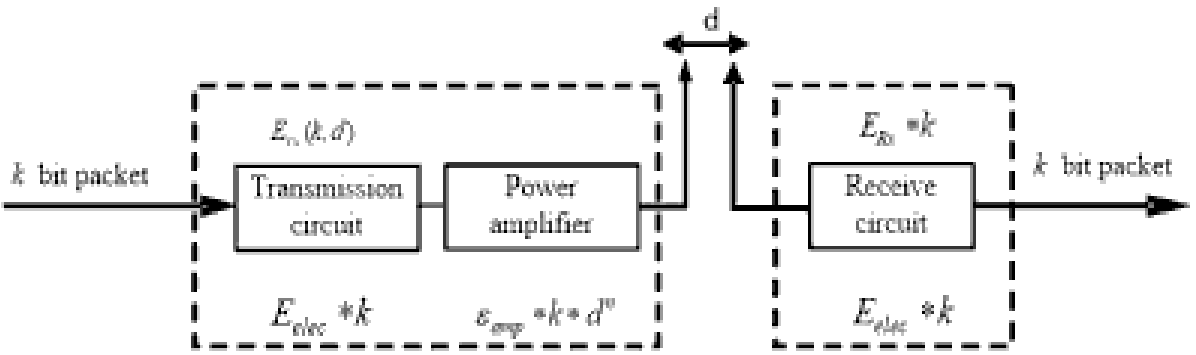


Figure 3.1. Radio model

$$ETx(k,d) = Eelec*k + Eamp*k*d*d \quad (1)$$

$$ERx(k) = Eelec*k \quad (2)$$

3.2 Direct Transmission Protocols

The direct transmission protocols consist of only base station and normal sensor nodes. The sensor nodes work as senders and base station serves as the destination node to all other sensor nodes in the network. The sensor nodes transmit their data to the base station and the base station received all data.[1] The senor nodes and base station is directly communicated without any intermediate communicator. The nodes only remain active during the data transmission to the base station. Since the nodes do not spend energy when receiving the messages for the other nodes, they will spend the minimum data on listening the channel and then they will spend their battery capacity on sending messages to the base station [1]. If a

sensor node transmits data directly to the base station, the energy loss incurred depends on the location of the base station; the energy loss incurred depends on the location of the sensor nodes relative to the base station. As a result, the direct transmission protocol's complexity can be ignored, but it is the least energy efficient protocol in most cases [2].

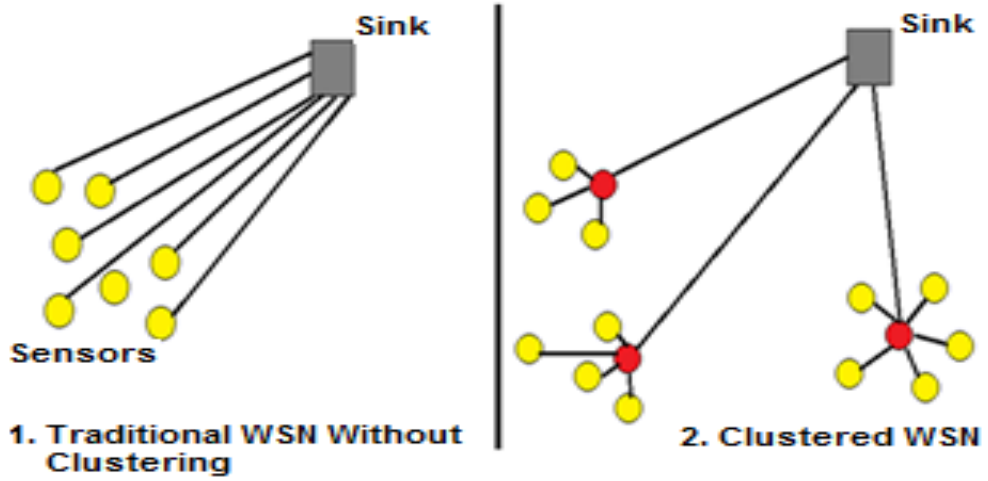


Figure 3.2.Architecture of Direct Transmission Protocol

3.3 LEACH Protocol

(Heinzelman, W.R, et al 2000) Suggested a new adaptive clustering algorithm using distribution algorithm for the nodes that organize themselves into clusters shown in figure (3.2). The main concept of LEACH is to create clusters of sensor nodes based on the strength of the received signals and use the cluster heads as routers to the base station. Since data communication to the station is the primary source of the energy consumption, of the cluster heads rotate among the sensor nodes. This idea treats the problem of the present of the cluster head during the network lifetime [8].

The operation LEACH is divided into rounds in which each round contains two phases:

Setup phase.

Steady state phase.

In the setup phase, the clusters are arranged and cluster-heads are chosen. In the first round, each node selects a random number between 0 and 1 and compares it to the threshold T (n) given in (3.1) and if the number is less than a threshold, the node becomes a cluster head.

$$T(n) = \begin{cases} \dfrac{P}{1 - P * (r \bmod \frac{1}{P})} & if\ n \in G \\ 0 & otherwise \end{cases}$$

Where P is the desired percentage of cluster heads, r is the current round, and G is the set of nodes that have not been clusters heads in the last 1/P rounds. In each round, selected cluster-heads broadcast an advertisement message to all the nodes in the network, information their new status [4] [8].

After receiving this message, each of the non- cluster-head nodes can determine to which cluster they belong to based on the strength of the received signal. Then, according to the number of nodes in a given cluster, that cluster's – head generates a TDMA (Time Division Multiple Access) schedule, and broadcasts a transmission time window to its CHs [10]. TDMA is a scheme where all concerned earth stations use the same carrier frequency and bandwidth and bandwidth with time sharing, non-overlapping intervals [9].

In the steady state phase Nodes in each cluster can begin sensing the information and transmitting sensed information to their own cluster- head throughout the distributed transmission time. The cluster-head node conducts the data fusion, aggregating, compressing and then sending the aggregated data to the base station. Since the BS is usually far from the field, communicating to the base station will consume a lot of the cluster-heads energy [8]. When the designated transmission time is over, the steady state phase finishes and the network retreats into the setup stage and begins an alternate round, starting with choice of new cluster-heads [10].

3.4 Cluster Formation

LEACH possibly chooses too many cluster heads at a time or randomly selects the cluster heads far away from the BS without considering nodes residual energy. As a result, some cluster heads drain their energy early thus reducing the lifespan of WSN [17].

In each round of the cluster formation, network needs to follow the two steps to select cluster head and transfer the aggregated data.

1) Set –up phase, which is again subdivided in to Advertisement, cluster set-up and schedule creation phases.

2) Steady – State Phase, which provides data transmission using Time Division Multiple Access (TDMA).

3.5 Methods for increasing lifetime of Wireless Sensor Networks using LEACH followed by direct mode:

According to W.B Heinzelman**:** In Wireless sensor networks is sensor nodes is distributed in the field then use hierarchical routing protocol making a cluster then cluster members selection the cluster head and all cluster members send data packet to the cluster's heads. After that cluster heads collected the data from the cluster nodes and data compress and aggregated, then send to the base station. Another method of if nodes to a member of the cluster then independent to send data to the base station depends on distance and remain energy, joined another cluster and send data to another cluster head.Many observations on MATLAB Simulink run and plot the graph then find that the if rounds maximum taken 1000 and run the programmed-on MATLAB Simulink then observe that find out the first node is died out in 200 rounds and around 90% node died out 900 rounds. And probably means that network dead. Then find out different rounds 700, 750,800,850,900, and 950 results nodes remaining energy and distance, most the nodes alive 25, 22 ,20,15, 11. Then observe that if after 700 rounds network is sends data to the base station. Not followed the leach protocol. For direct communication find out data from leach protocol and find out the nodes remain energy of each nodes and distance and node id then find each node send data to the base station through table 3.1- 3.2.

3.6 LEACH followed by Direct Mode

Many rounds randomly apply to the networks and see that cluster are formed and nodes joined to the cluster heads and cluster head sends data to the base station. In randomly rounds nodes is member of cluster and another cluster node not member of the cluster. After a certain number of rounds when node density drops below a threshold value, no clustering of further node but all nodes transmit directly data packets to the base station.

Let **Threshold define of nodes = x no of nodes / area in square meters**

That is the observation Table.2 shows the nodes of density and remains energy of each nodes, and distance of each nodes for transfer data to base station required

Table 3.1. Maximum rounds 700 Each node send data to the BS.

Node id	Distance	Remains energy	Direct data send to the BS
32	144	0.0030	2.914
45	96	0.1215	2.132
52	83	0.1629	343
64	87	0.1834	364
81	140	0.0275	27
92	124	0.1820	295
112	35	0.2104	844
114	52	0.1915	634
160	82	0.1951	416
163	93	0.1926	352
187	68	0.1472	382
199	33	0.2670	1096
200	55	0.1430	445
116	78	0.1720	387
176	135	0.0709	76
191	27	0.2034	887
194	122	0.0849	106
189	77	0.1376	314
195	86	0.1552	313

188	59	0.1745	514
192	85	0.1506	307

Table no 3.2. Maximum rounds 800 total no of node alive

Node id	**Distance**	**Remains energy**	**Direct data send to the BS**
3	100	0.0831	138
23	84	0.0994	206
35	55	0.1388	432
56	61	0.1568	327
70	72	0.1139	279
80	43	0.2040	747
93	107	0.0321	201
121	74	0.1323	178
124	109	0.0746	79
129	112	0.0539	186
146	45	0.1309	465
163	93	0.1479	271
164	76	0.1126	261
179	59	0.2450	339
189	77	0.0840	722
190	101	0.0859	192
192	85	0.1030	141
200	55	0.1001	210
199	33	0.2345	311
189	77	0.0540	192
188	59	0.1303	247
195	86	0.1028	207
190	101	0.0859	141

Above table shows that each node of remains energy, distance. In above find out for each node transmitting energy of each nodes for transmitting the energy to the base station.

After maximum rounds 900 each node energy is very low and no of alive node low. After a certain no of rounds when nodes density drops below a threshold value, no clustering a further done but all nodes remains alive possibility to send data to the base station directly without LEACH protocol.

After that a certain round 700 observe that the remains energy and distance is efficient for the direct communication to the BS. Around 25 nodes alive at rounds 700 and remains energy is efficient for transmitting data to BS.

Let **threshold density of nodes = no nodes alive/ area in square meter.**

In above table check out the density of nodes if n > threshold value then follows the leach protocol.

If threshold value n< threshold value then the do not perform clustering, all nodes transmit data packet direct to the base station.

Formulae for transmitting the energy of each nodes to radio model

Etx (k,d)= Eelec*k+ Eamp *k *d*d

Etx (k,d) = k (Eelec + Eamp*d*d)

Where k= data packets assume 4000, d= distance of node to the BS, Eelec = 50 nJ/bit, Eamp = 100 pJ/bit/m*m

This is the transmitting energy formula for find each node transmit energy.

Another formula for transmitting energy to the base station in direct mode communication

No of packtets that could be sent directly to the base station = node remains energy / transmitting energy Etx

In above table 2 and 3 find out each node when maximum rounds are increase the nodes energy is decrease and quickly died out.

After 700 rounds then network is communicated to the base station of each node then each node transmits energy is more than the leach protocol.

We calculate the number of messages sent to the direct mode to the base satiation our results show that better energy efficiency and increase the lifetime of the network each node given better performance.

Compare to the LEACH protocol the direct mode communication is more efficient energy and increase the lifetime of the network.

In leach protocol 90% nodes died out 950 rounds then understand that network is died out. Because no of node is equal to the cluster head, each node is itself to cluster head, in clustering the energy is required so this main reason for died out the network.

In direct mode one node is alive the network performs better to the leach. Because no clustering formation. Each node communicates to the BS. In single hop mode.

4 SIMULATION & RESULTS

4.1 Simulation Tool

The simulation tool has been carried out on MATLAB R2018a software on the basis of the simulation; LEACH more than doubles the useful network life as shown in Figure below. To read more about the effects of the simulation. Although LEACH improved the life of the network and dissipated energy, it does have some disadvantages. LEACH assumes, for example, that the initial energy occurs at all nodes, but does not occur in real time. In addition, it implies that nodes are static. For mobile nodes, it is also difficult to submit. In addition, the failure of CHs produces other issues and does not take account of the current multiple BS. Finally, the dynamical clustering has a further overhead. Many protocols were found to improve and enhance LEACH according to these disadvantages.

4.2 Experimental Settings

In this dissertation/project compare between two mode communications LEACH protocol and leach followed by direct mode. MATLAB R2018A used in the experiments. Table 4.1 summarizes the experimental setting used. These values were selected based on some tuning that was done until values were reached that could be read, recorded, and analyzed.

Assume a number of 200 sensor nodes distributed in field 300*300 meters. Nodes were placed at random points, each had (x, y) coordinates. The BS (sink) was located at the sensor field (50, 150)

Table 4.1. Summary of parameter settings.

Parameters	**Values**
Number of nodes	200
Sensor field area	300*300

Maximum rounds	1000
Sink	50*150
Initial Node Energy	50 nJ/bit
CH probability	0.1
Etx	50nJ/bit
Erx	50nJ/bit
Data aggregation Eda	5nJ/bit/message
Free space amplifier parameters Efs	10pJ/bit/m^2
Multipath amplifier parameter Emp	0.0013 pJ/bit/message
Channel type	Wireless
Communication channel	Bidirectional

Basic operation of both protocols assumes having CHs and CMs, these are sometimes referred to as advanced and normal nodes respectively, the total number of nodes in the WSN is equal to the number of sensor nodes plus one sink, the initial energy in each node 0.2 nJ/bit. As basic operation of LEACH selects a node to be CH based on a value between 0 and 1.The experiment of the following values was set: Etx = Erx = 50nJ/bit, Efs = 10 pJ/bit/m^2, Emp = 0.0013 pJ/bit/m^4 and Eda = 5nJ/bit/message. The simulation run for 1000 rounds.

4.3 RESULTS

In the following pictures figures line is show active nodes in the LEACH protocol randomly 0.5J. Horizontal coordinates expresses the simulation time (seconds). Vertical coordinates express the active node number left. The simulation result shows that when the active nodes of the LEACH protocol retain 40% the active nodes of the LEACH followed by direct mode just start decrease. Randomly nodes with initial energy of 1J. the simulation result shows that the leach followed by direct mode has a more pronounced effect.

Therefore, the leach followed by direct mode can average energy consumption and effectively prolong the network lifetime. The reason is flowing on the one hand, the leach followed by direct mode takes the energy factor in to consideration. The threshold is different from each node. The node with more remaining energy has more opportunities to become a cluster head. On the other hand, every cluster head has more than one routes to send the data to the sink, cluster heads can select the optimal path to transmit information and reduce the of energy of cluster head. We can see the leach followed by direct mode can average energy consumption of every node and dispersed the network flow. The life cycle of the network is extended.

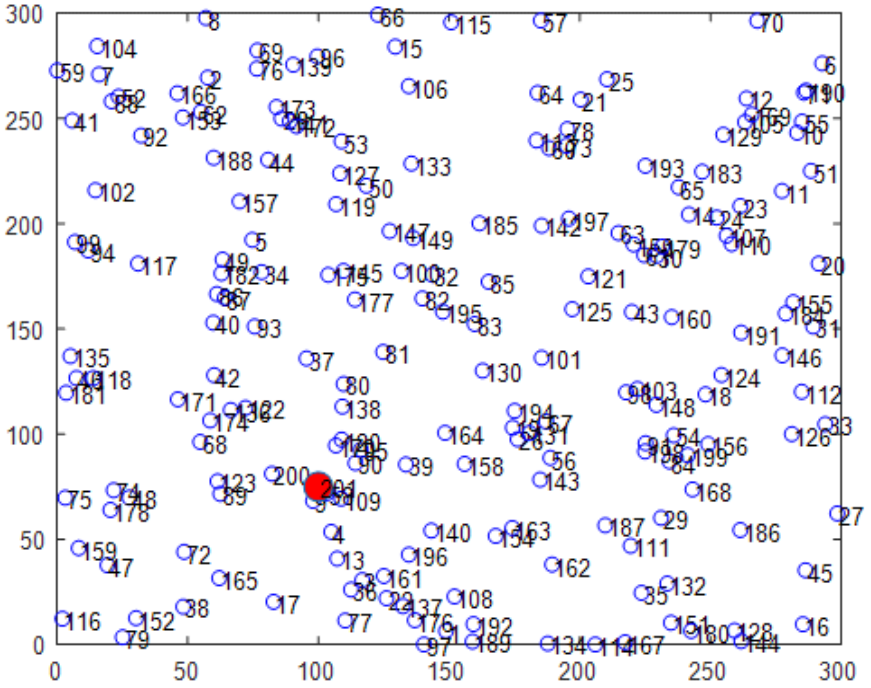


Figure 4.1. Distribute Nodes (x, y)/location of BS

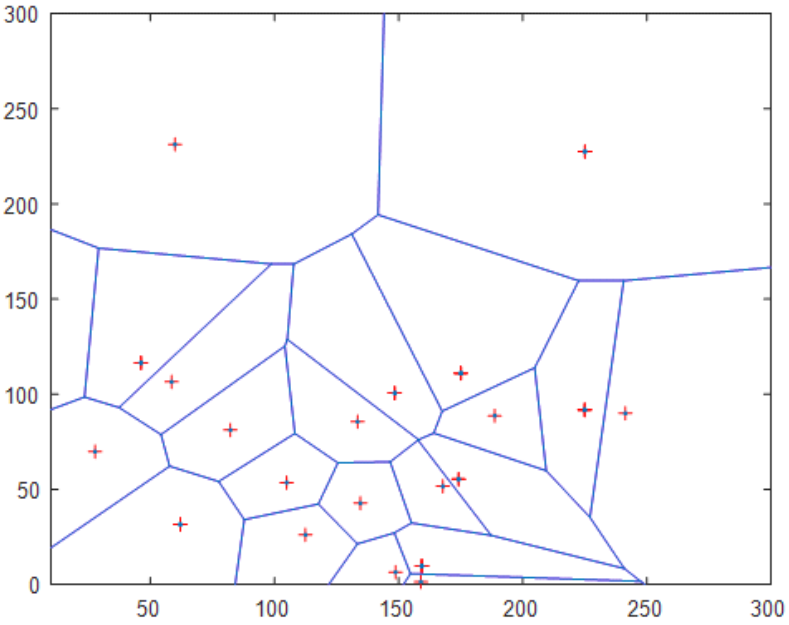


Figure 4.2. Location of CHs

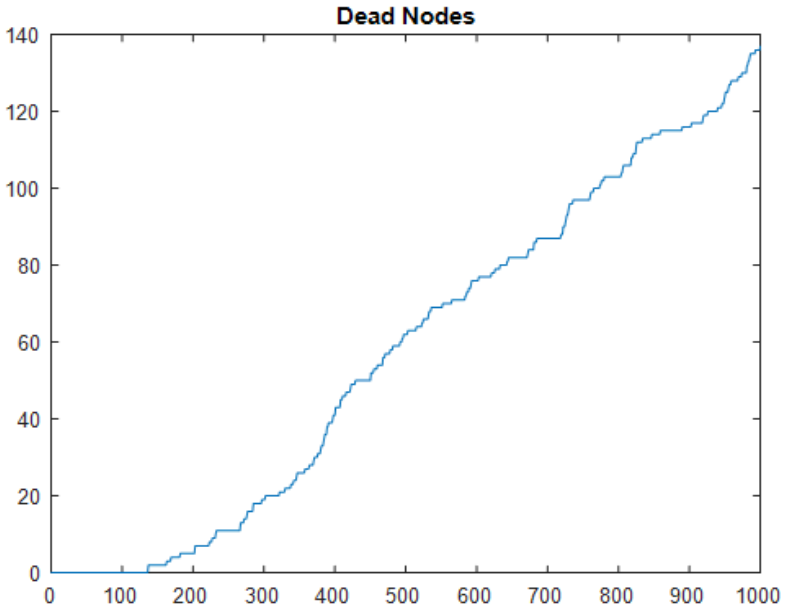


Figure 4.3. No. of dead node/round

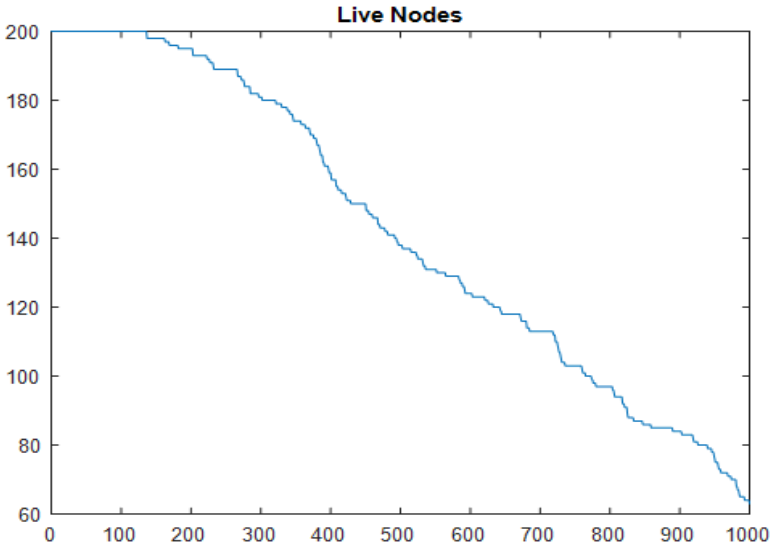


Figure4.4 No. of alive nodes/round

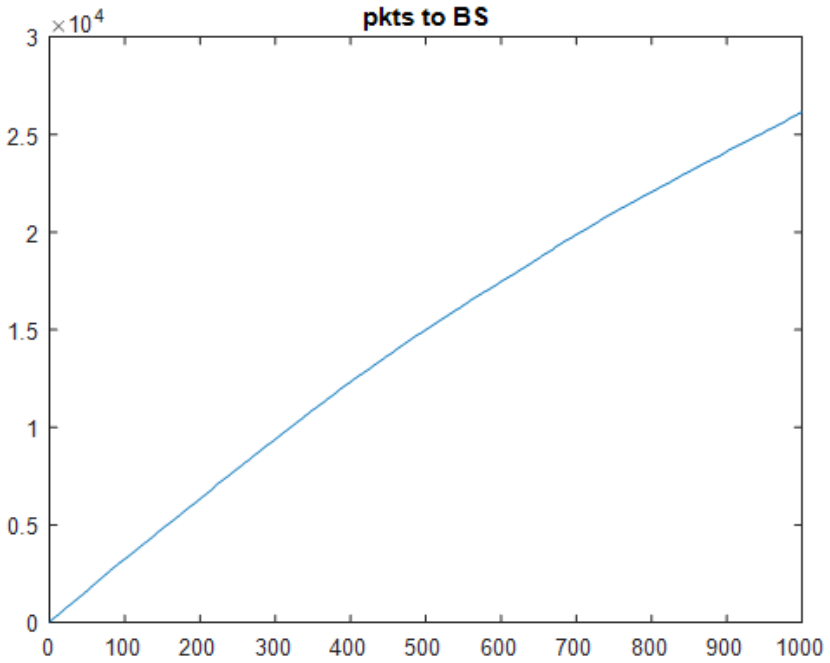


Figure 4.5. Packet to BS Per round/max. round

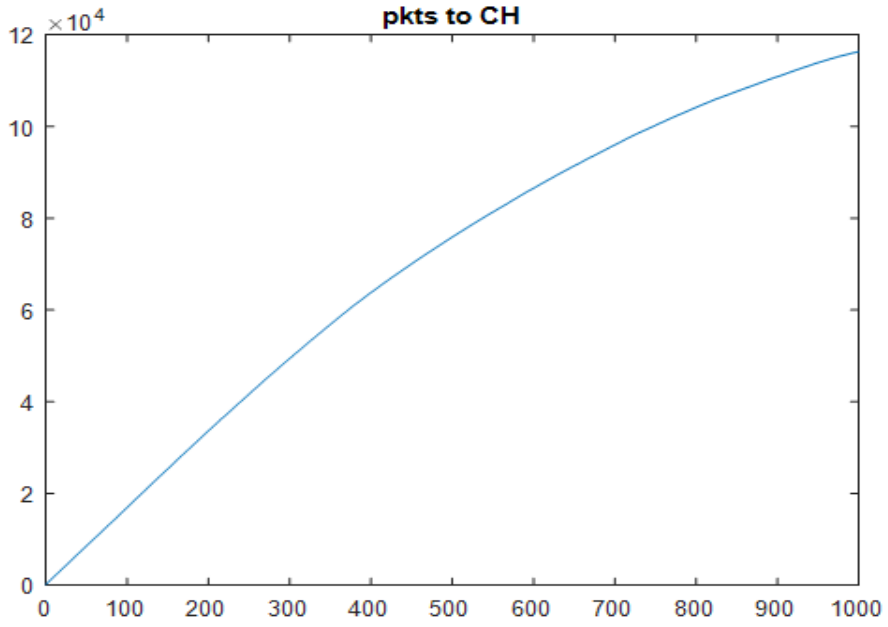


Figure 4.6. Packet to CH/MAX, round

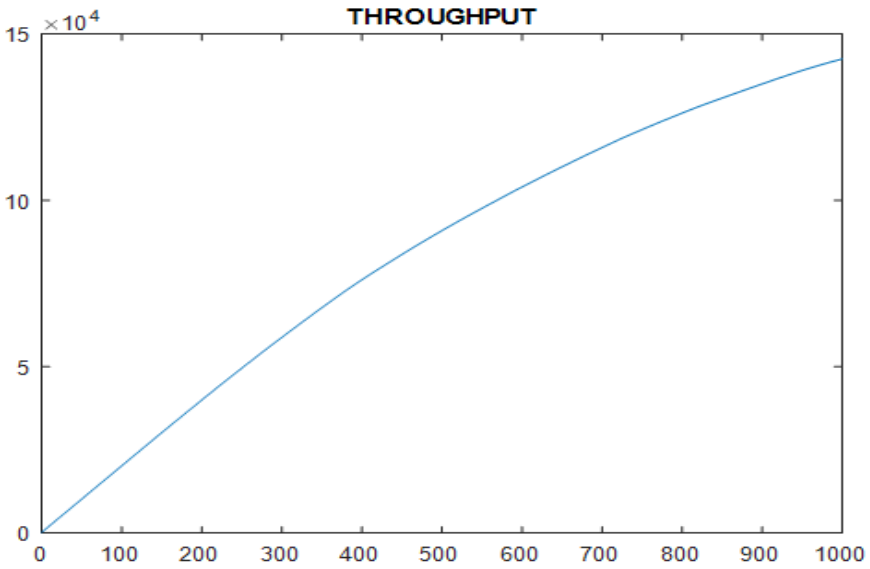


Figure 4.7. Throughput/max. round

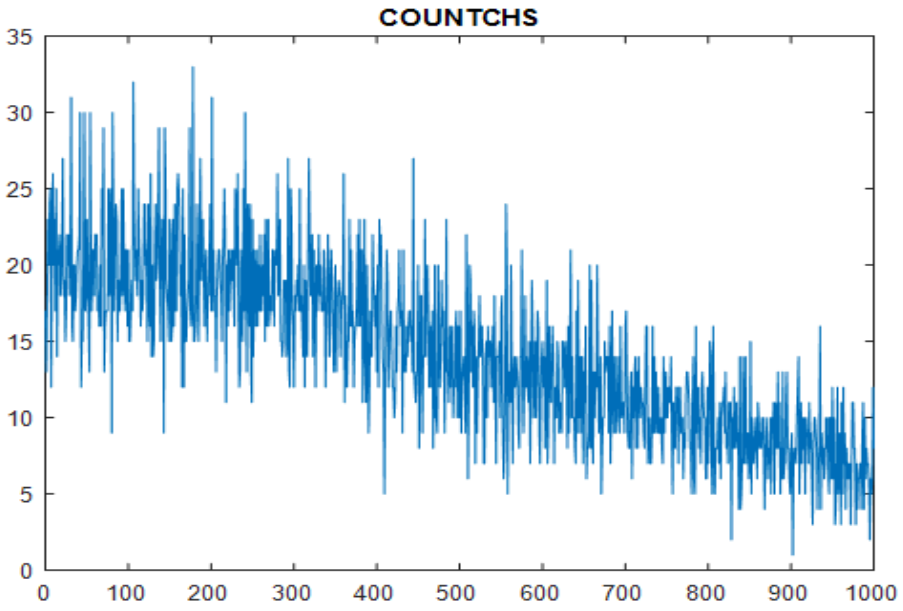


Figure 4.8. Count CHs/max round

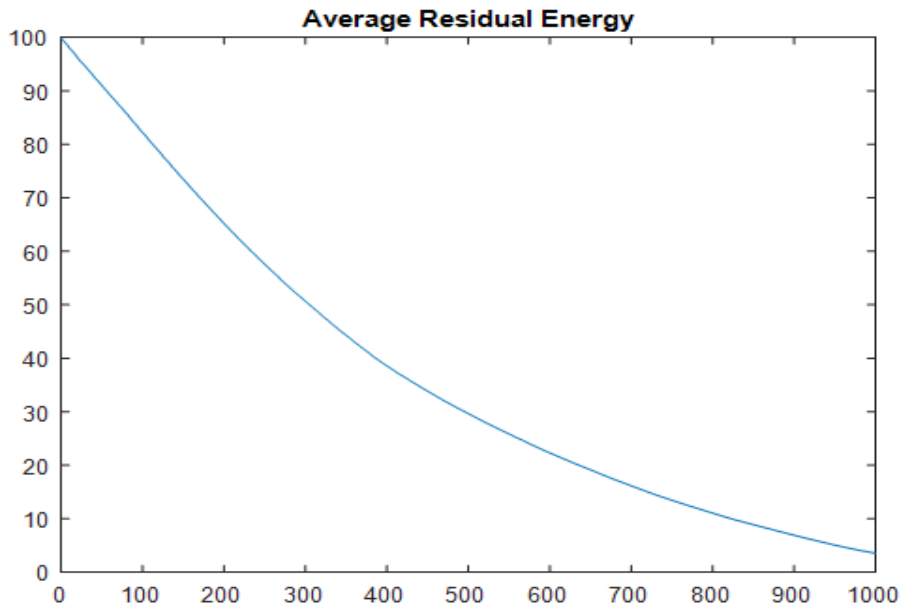


Figure 4.9.Average residual energy /round

Our results/observation shows that direct mode communication to the base station is better energy efficient, reliability and network scalability lastly, we summarize the results in table 4.2 by calculating the first node died and last node alive for the protocol.

Table 4.2. FND, LND FOR LEACH FOLLOWED BY DIRECT MODE

METRICS	LEACH	LEACH FOLLOWED BY DIRECT
FIRST NODE DIED	150	850
Last node died	After 950	After 1500

The Figure below shows alive nodes versus number of rounds plot for original LEACH and LEACH followed by Direct mode. The FND (First Node Dead) also known as stability period for LEACH protocol is at 390 round means that the network is stable up 390 rounds in LEACH protocol, the number of round increases in LEACH protocol then find out after 900 rounds probably 90 percent of node is die out in the networks. After a certain round find out the networks and observe that maximum rounds 700, 800 and 900 mostly node is alive is

20, 15 and 10. then after a certain round decide that leach followed by direct mode works in after 700 rounds in networks.In leach followed by direct mode first node dead is 790 round, and last nodes die out after 1500 round.That means leach followed by direct mode work better than an original LEACH protocol and our network will stable till 1500 round and all the nodes are alive which shows improvement in leach protocol.

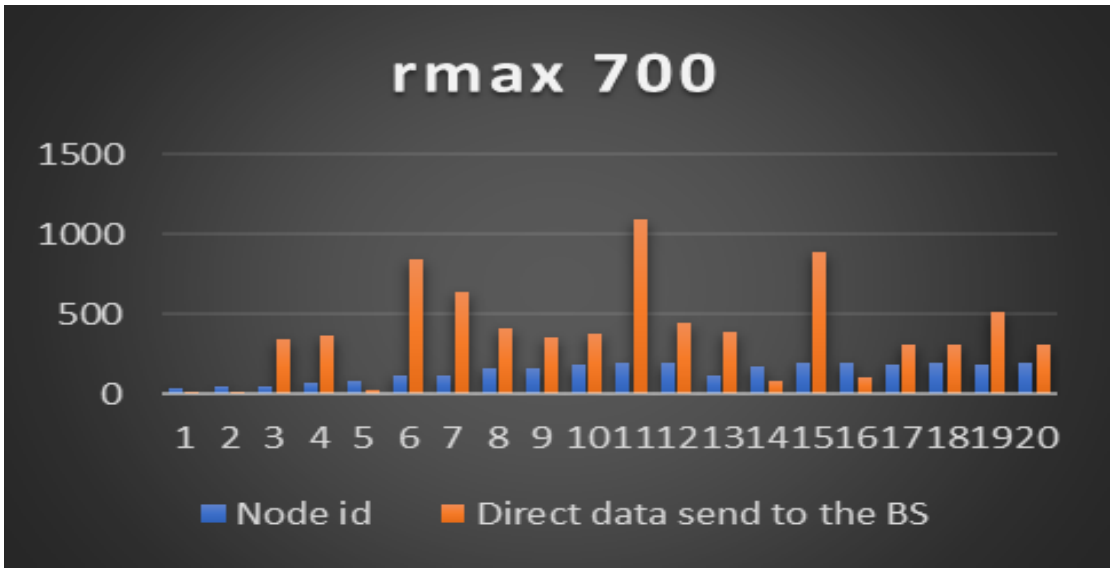


Figure 13 LEACH followed by direct communication to the BS after 700 rounds

Figure 4 shows that nodes in leach followed by direct mode and leach die out earlier. In leach and leach followed by direct mode almost all the nodes have dead before reach to the last round but in leach followed by direct mode some node that are alive.

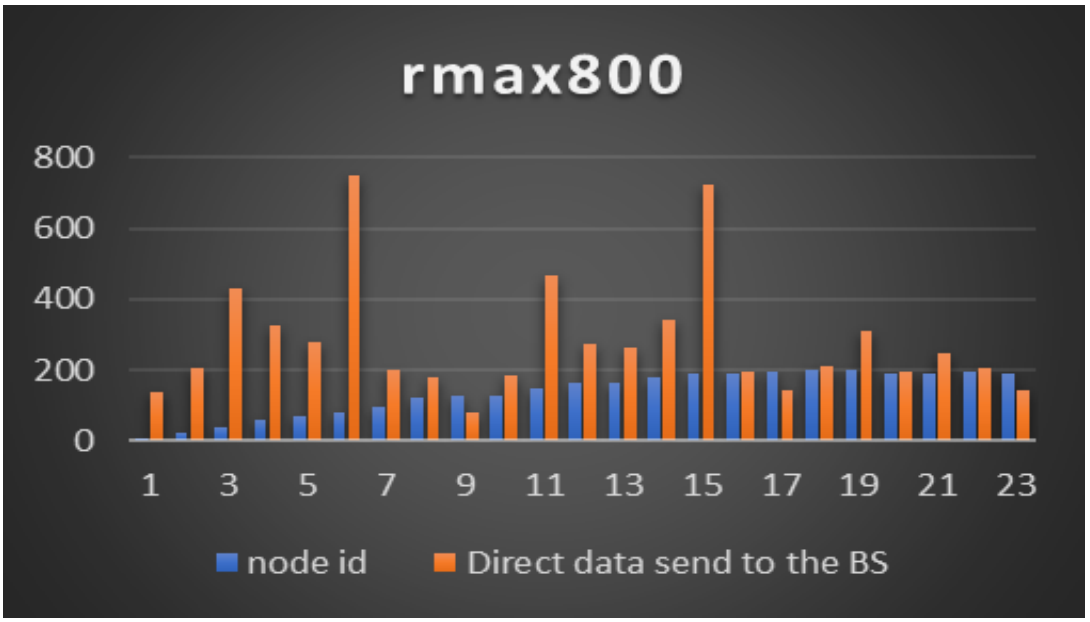


Figure 14. LEACH followed by direct communication to the BS after 800 rounds

CONCLUSION

In this protocol for wireless sensor networks called leach protocol which is the first and the most important protocol in wireless sensor networks which is used cluster-based transmission of energy. Followed by overview LEACH protocol implementations, and then consider the new version of LEACH followed by the direct communication called direct mode communication in wireless sensor networks. From the simulations results seen the in-LEACH protocol performs maximum 900 rounds and after that dead networks because some nodes are itself a cluster heads, the second LEACH followed by direct mode is start the around 700 and performs the around round 1500. Because when one node is alive it performs till the node is dead out. That is proving that in table 4. That means the new version of LEACH followed by direct mode outperforms the original version of LEACH protocol

REFERENCES

[1] Lindsey et al. G.S. (2013) came about the proposition of PEGASIS which is an extension of LEACH, International Conference on IEEE, -41-45.

[2] Alkhatib, A.A.A., & Baicher, G.S. (2012). Wireless sensor network Architecture. International conference on Computer Networks and Communication systerms,35, 11-15.

[3] [Aslam, M., et al. 2012]). A Survey on routing protocols for wireless sensors networks. Ad hoc networks, 3, 325-349.

[4] Banerjee, T., et al. (2013) L-LEACH: An improved LEACH protocol for lifetime improvement in WSN. International Journal of Engineering Trends and Technology, 4,1521-1526.

[5] Behrouz, R. (2002) Increasing network lifetime in cluster based WSNs Via Fuzzy Logic. Toronto: Ryerson University.

[6] Dakshayini, M., & HS, P. K. G. (2013). Energy Aware Dynamic Clustering and Hierarchical Routing based on LEACH for WSN. International of Computer networking, wireless and Mobile communication, 3(3), 79-86.

[7] Gupta, A., Goyal, M., & Malik, S. (2015). Clustering Approach for Enhancing Network Energy using LEACH protocol in WSNs. International Journal of Wired and Wireless communications, 2(1), 20-25

[8] Heinzelman, W. R., et al. (2000). Energy – efficient communication protocol for wireless microsensor networks. International Journal of system sciences, 2(11), 1-10.

[9] Sharma, M., & shaw, A. K, (2012). Transmission Time and Throughput analysis of EEE LEACH, LEACH and Direct Transmission Protocol: A Simulation Base Approach, 3(5). 97-104.

[10] Shu. W, et al. (2017). An optimized multi-hop routing algorithm base on clonal selection strategy for energy-efficient management in wireless sensor networks. 22(7),8-14.

[11] W.R. Heinzelman, A. Chandrakasan, and H. Balakrishanan, Energy Efficient communication protocol for wireless micro sensor networks, in proceeding of the 33rd Hawaii interanational conference on syterm science vol .8. pp8020. 2000.

[12] M. Bhardwaj and A.P. Chandrakasan, Bounding the lifetime of sensor networks via optimal role assignment, in the proceeding IEEE INFOCOM, vol, 3, pp 1587-1596. 2002.

[13] Wei Cheng proposed a novel adaptive, distributed, energy efficient clustering algorithm, AEEC for wireless sensor network, in the proceeding IEEE INFOCOM, vol, 3, pp 87-96. 2012.

[14] Heidemann and Estrin, flooding in order to minimize energy consumption in wireless sensor network, International Conference on pp 93-98. nov, .2015.

[215] Abdullah, M. Y., et al. (2009). Energy scheduling with roles dormant cells in wireless sensor network. In Mechatronics and Autamation, International Conference on IEEE, -541-542.

[16] Gauram, G. & B., Design and Simulation of wireless sensor network in NS2, International Journal of computer applications, pp 14-16,2015.

Matlab code of LEACH Protocol

```
clear all;
close all;
rng (&#39;Default&#39;);
x=300;            %distance measure in meter
y=300;            %distance measure in meter
sink.x=0.5*x;
sink.y=0.5*y;
sink.x=100;
sink.y=75;
n=200;
p=0.1;
E=0.5;
ETX=50*0.000000001;
ERX=50*0.000000001;
Efs=10e-12;
Emp=0.0013e-12;
EDA=5*0.000000001;
r max=950;
do=sqrt(Efs/Emp);
Et=0;
%%%
```

```
S(i).yd=rand(1,1)*ym;
YR(i)=S(i).yd;
distance=sqrt( (S(i).xd-(S(n+1).xd) )^2 + (S(i).yd-(S(n+1).yd) )^2 );
S(i).distance=distance;
S(i).G=0;
%initially there are no cluster heads only nodes
S(i).type=&#39;N&#39;;
S(i).E=Eo;
Et=Et+S(i).E;
figure(h*10)
plot(S(i).xd,S(i).yd,&#39;bo&#39;);
text(S(i).xd+1,S(i).yd-0.5,num2str(i));
hold on;
end
plot(S(n+1).xd,S(n+1).yd,&#39;o&#39;, &#39;MarkerSize&#39;,
12, &#39;MarkerFaceColor&#39;, &#39;r&#39;);
text(S(n+1).xd+1,S(n+1).yd-0.5,num2str(n+1));
hold off ;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
countCHs=0; %variable, counts the cluster head
cluster=1; %cluster is initialized as 1
flag_first_dead=0; %flag tells the first node dead
flag_half_dead=0; %flag tells the 10th node dead
flag_all_dead=0; %flag tells all nodes dead
first_dead=0;
half_dead=0;
all_dead=0;
allive=n;
%counter for bit transmitted to Bases Station and to Cluster Heads
packets_TO_BS=0;
packets_TO_CH=0;
packets_TO_BS_per_round=0;
```

```
%%%
```

```
if(flag_half_dead==0)
half_dead=r;
flag_half_dead=1;
end
end
if(dead==n)
if(flag_all_dead==0)
all_dead=r;
flag_all_dead=1;
end
end
%hold on;
end
if S(i).E&gt;0
S(i).type=&#39;N&#39;;
end
end
%plot(S(n+1).xd,S(n+1).yd,&#39;x&#39;);
STATISTICS.DEAD(h,r+1)=dead;
STATISTICS.ALLIVE(h,r+1)=allive-dead;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
countCHs=0;
cluster=1;
for i=1:1:n
if(S(i).E&gt;0)
temp_rand=rand;
if ( (S(i).G)&lt;=0)
%Election of Cluster Heads for normal nodes
if ( temp_rand &lt;= ( p/ ( 1 - p * mod(r,round(1/p)) )) )
countCHs=countCHs+1;
packets_TO_BS=packets_TO_BS+1;
packets_TO_BS_per_round=packets_TO_BS_per_round+1;
PACKETS_TO_BS(r+1)=packets_TO_BS;
```

```
S(i).type=&#39;C&#39;;
S(i).G=round(1/p)-1;
C(cluster).xd=S(i).xd;

C(cluster).yd=S(i).yd;
%plot(S(i).xd,S(i).yd,&#39;k*&#39;);
distance=sqrt( (S(i).xd-(S(n+1).xd) )^2 + (S(i).yd-(S(n+1).yd) )^2 );
C(cluster).distance=distance;
C(cluster).id=i;
X(cluster)=S(i).xd;
Y(cluster)=S(i).yd;
cluster=cluster+1;
%Calculation of Energy dissipated
distance;
if (distance&gt;do)
S(i).E=S(i).E- ( (ETX+EDA)*(4000) + Emp*4000*(
distance*distance*distance*distance ));
end
if (distance&lt;=do)
S(i).E=S(i).E- ( (ETX+EDA)*(4000) + Efs*4000*( distance * distance ));
end
end
end
end
end
STATISTICS.COUNTCHS(h,r+1)=countCHs;
% or STATISTICS.COUNTCHS(h,r+1)=clster-1;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%Election of Associated Cluster Head for Normal Nodes
for i=1:1:n
if ( S(i).type==&#39;N&#39; &amp;&amp; S(i).E&gt;0 )
```

```
if(cluster-1&gt;=1)
min_dis=sqrt( (S(i).xd-S(n+1).xd)^2 + (S(i).yd-S(n+1).yd)^2 );
min_dis_cluster=0;
for c=1:1:cluster-1
temp=min(min_dis,sqrt( (S(i).xd-C(c).xd)^2 + (S(i).yd-C(c).yd)^2 ) );
if ( temp&lt;min_dis )
min_dis=temp;
min_dis_cluster=c;
end
end
%Calculating the culsterheads%
if(min_dis_cluster~=0)
min_dis;
if (min_dis&gt;do)
S(i).E=S(i).E- ( ETX*(4000) + Emp*4000*( min_dis * min_dis * min_dis *
min_dis));
end
if (min_dis&lt;=do)

S(i).E=S(i).E- ( ETX*(4000) + Efs*4000*( min_dis * min_dis));
end
S(C(min_dis_cluster).id).E = S(C(min_dis_cluster).id).E- ( (ERX + EDA)*4000 );
packets_TO_CH=packets_TO_CH+1;
else
min_dis;
if (min_dis&gt;do)
S(i).E=S(i).E- ( ETX*(4000) + Emp*4000*( min_dis * min_dis * min_dis *
min_dis));
end
if (min_dis&lt;=do)
S(i).E=S(i).E- ( ETX*(4000) + Efs*4000*( min_dis * min_dis));
end
packets_TO_BS=packets_TO_BS+1;
```

```
packets_TO_BS_per_round=packets_TO_BS_per_round+1;
PACKETS_TO_BS(r+1)=packets_TO_BS;
end
S(i).min_dis=min_dis;
S(i).min_dis_cluster=min_dis_cluster;
else
min_dis=sqrt( (S(i).xd-S(n+1).xd)^2 + (S(i).yd-S(n+1).yd)^2 );
if (min_dis&gt;do)
S(i).E=S(i).E- ( ETX*(4000) + Emp*4000*( min_dis * min_dis * min_dis *
min_dis));
end
if (min_dis&lt;=do)
S(i).E=S(i).E- ( ETX*(4000) + Efs*4000*( min_dis * min_dis));
end
packets_TO_BS=packets_TO_BS+1;
packets_TO_BS_per_round=packets_TO_BS_per_round+1;
end
end
end
STATISTICS.PACKETS_TO_CH(h,r+1)=packets_TO_CH;
STATISTICS.PACKETS_TO_BS(h,r+1)=packets_TO_BS;
STATISTICS.PACKETS_TO_BS_PER_ROUND(h,r+1)=packets_TO_BS_per_round;
STATISTICS.THROUGHPUT(h,r+1)=STATISTICS.PACKETS_TO_BS(h,r+1)+STATISTI
C
S.PACKETS_TO_CH(h,r+1);
En=0;
for i=1:n
if S(i).E&lt;=0
continue;
end
En=En+S(i).E;
end
ENERGY(r+1)=En;
```

```
STATISTICS.ENERGY(h,r+1)=En;

end
first_dead_LEACH(h)=first_dead
half_dead_LEACH(h)=half_dead
all_dead_LEACH(h)=all_dead
% cluster head display-------
figure(11)
warning(&#39;OFF&#39;);
[vx,vy]=voronoi(X(:),Y(:));
plot(X,Y,&#39;r+&#39;,vx,vy,&#39;m-&#39;);
hold on;
voronoi(X,Y);
axis([10 xm 0 ym]);
end
for r=0:rmax
STATISTICS.DEAD(h+1,r+1)=sum(STATISTICS.DEAD(:,r+1))/h;
STATISTICS.ALLIVE(h+1,r+1)=sum(STATISTICS.ALLIVE(:,r+1))/h;
STATISTICS.PACKETS_TO_CH(h+1,r+1)=sum(STATISTICS.PACKETS_TO_CH(:,r+1))/
h
;
STATISTICS.PACKETS_TO_BS(h+1,r+1)=sum(STATISTICS.PACKETS_TO_BS(:,r+1))/h
;
STATISTICS.PACKETS_TO_BS_PER_ROUND(h+1,r+1)=sum(STATISTICS.PACKETS_
T
O_BS_PER_ROUND(:,r+1))/h;
STATISTICS.THROUGHPUT(h+1,r+1)=sum(STATISTICS.THROUGHPUT(:,r+1))/h;
STATISTICS.COUNTCHS(h+1,r+1)=sum(STATISTICS.COUNTCHS(:,r+1))/h;
STATISTICS.ENERGY(h+1,r+1)=sum(STATISTICS.ENERGY(:,r+1))/h;
end
first_dead=sum(first_dead_LEACH)/h;
half_dead=sum(half_dead_LEACH)/h;
all_dead=sum(all_dead_LEACH)/h;
```

```
r=0:rmax;
figure(1)
plot(r,STATISTICS.DEAD(h+1,r+1));
title(&amp;#39;Dead Nodes&amp;#39;)
figure(12)
plot(r,STATISTICS.ALLIVE(h+1,r+1));
title(&amp;#39;Live Nodes&amp;#39;)
figure(3)
plot(r,STATISTICS.PACKETS_TO_BS(h+1,r+1));
title(&amp;#39;pkts to BS&amp;#39;)
figure(4)
plot(r,STATISTICS.PACKETS_TO_BS_PER_ROUND(h+1,r+1));
title(&amp;#39;pkts to BS per round&amp;#39;)
figure(5)
plot(r,STATISTICS.PACKETS_TO_CH(h+1,r+1));
```